Christopher Redfearn

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SUMMARY

- Over 10+ years of experience as a Hardware and System Test Engineer.
- Developed Python web applications for identifying defects using manufacturing inspection images.
- Developed Python automation scripts for measuring hardware, resulting in 15% increase in efficiency.
- Validated and verified performance, functionality, and quality specs for prototypes.
- Performed reliability testing on electrical and optical modules for complete lifetime testing in thermal chambers.
- Conducted manual testing and validation to ensure the accuracy and reliability of the python applications.
- Defined test plan, implemented test setup/processes, including design of test fixtures, test equipment and test software.
- Provided root cause analysis/failure analysis for reworked/new prototypes, solved issues at component and board level.
- Tested PCBs, electrical components, cameras, lasers and optical elements.

PROFESSIONAL EXPERIENCE

QA Engineer III (Performance)
Iconma for Confidential Client (FAANG), Sunnyvale, CA **May 2025 – Present**

- Performed electrical validation on consumer tablets to characterize touch sensor and impedance behavior under multiple charger configurations
- Utilized impedance analyzers and Python-based scripts to collect and process test data for signal quality and accessory compatibility

Hardware Display Test Engineer
Basic Solutions for Meta, Sunnyvale, CA **Dec 2024 - Jan 2025**

- Spearheaded the development and implementation of the streamlined inventory tracking system, increasing accuracy and improving asset management efficiency by 20%.
- Operated and maintained advanced lab test equipment, including function generators, spectrum analyzers, thermal chambers, and specialized optical test instruments (e.g., photospectrometers) for validating display performance and accuracy.
- Acting Lab Manager and led a group of 3 to delegate daily tasks around testing and lab equipment needs.

Senior Engineer
Apex Systems for Cruise, San Francisco, CA **Jul 2024 - Nov 2024**

- Managed electrical engineering testing on sensors including Lidar and cameras and delegated lab space to teams
- Tested audio sensors in thermal chambers and anechoic chambers and collected quality data
- Assisted with off-site testing for data collection of long range lidars and ground truth testing with Leica measuring device
- Acting Lab Manager and managed millions dollars worth of lab assets for tracking and calibration needs

Optical Engineer II
Mindlance for Meta , Sunnyvale, CA **Jan 2022 - Jan 2024**

- Developed Python applications and automation scripts to optimize hardware testing and data collection processes.
- Authored and executed detailed automation test protocols and test plans for sensor systems.
- Integrated Python apps into internal ground truth testing and external manufacturing for OLED and LCD displays.
- Conducted reliability testing on LCD and OLED displays for accelerated life testing.
- Managed lab inventory of devices, parts, test equipment and other supplies.
- Conducted inventory audit, verify device locations and inventory cycle counts for on-campus labs.
- Partnered with QA teams to plan and procure lab consumables, modules and test devices.

Test Technician
Johnson & Johnson, Santa Clara, CA **Jan 2021 - Jan 2022**

- Responsible for bring-up, testing, troubleshooting PCBAs, and verifying firmware for bench level PCBA testing.
- Expert in software tools related to Linux servers using CLI, shell commands, and SSH management.
- Generated reports on root cause analysis at component and board level for reworking boards.
- Validated automated tests for robotics, sensors, and complex med-device systems; and constructed, verified functionality and quality of motors, designed test fixtures and test setups.

Optical Lab Technician
Accenture for Facebook, Menlo Park, CA **Jun 2018 - Aug 2020**

- Defined test plan, implemented test setup and processes, including designing of test fixtures, software, and equipment.
- Responsible for bring-up of prototypes with electrical testing methods and verified firmware; tested PCBs, electrical components, cameras, lasers and optical elements.
- MATLAB development of data acquisition applications on test setups linked to network analyzers, spectrum analyzers, oscilloscopes and power meters.

Test Technician
West Valley Staffing for Molecular Devices, San Jose, CA **Oct 2017 - Apr 2018**

- Tested and validated robotics, lasers, sensors, and DNA sequencing platforms.
- Managed team communication of 3-4 testing electro-mechanical systems.

EDUCATION

M.S. Computer Science	Arizona State University, AZ	Projected May 2026
B.S. Electrical Engineering	San Jose State University, CA	Completed Dec 2015

SKILLS

Programming	Python, C++, MATLAB
Hardware	Function Generator, Soldering, Optical Power Meter, Spectrum Analyzer, Impedance Analyzers, Electronic Load, Photospectrometers, SEM, X-Ray, Microscopes
Software	Altium, OrCAD, Allegro, Arduino IDE, Beam Gauge, Solidworks, Word, Excel, PowerPoint, Radiant, TrueView
Lab	Class 1000 clean-room and laser safety protocols

VOLUNTEERING EXP.

Key Volunteer
Local non-profit partnered with city office, Our City Forest, San Jose, CA **Mar 2024 – Jun 2024**

- Key member in the initiative for acclimating nursery trees and building structures for support and resulted in an offer to become part of the team as a paid staff

Lead Volunteer
Loaves & Fishes Family Kitchen at Goodwill, San Jose, CA **Jul 2017 - Oct 2017**

- Tasked to lead a group of 10-15 volunteers in preparing and serving meal orders

MORE EXPERIENCE

Application Engineer
Camtek, Fremont, CA **Jul 2016 - May 2017**

- Proficient in optimizing performance of all Camtek inspection tools.
- Performed field installations and repairs in PPE in semiconductor fabs.
- Tested client samples and prepared results in presentation reports.
- Managed technical questions and requests for service/support visits.
- Acted as a communication highway for USA parts, USA clients, USA engineers, administrators, managers, HQ parts and HQ development teams.
- Calibrated 2D and 3D inspection systems/applications.
- Passed Basic Training/Safety course at international HQ (Israel).
- Communicated with up to 6-8 USA clients for managing.
- Traveled 80% time (Domestic and International).

PDM Analyst
Omega Solutions for Google, Mountain View, CA **Feb 2016 - Jul 2016**

- Created ECOs and BOMs, submitted changes, edited vendor sourcing, created items and added files.
- Created Google part numbers (GPN's) for new parts for Google Cloud and Consumer Hardware teams.
- Fulfilled phases of data cleaning and migration to Agile from Windchill database.
- Used advanced Excel tools for cleaning data with 100, 000+ lines.
- Attended daily "huddles" with large team to discuss project progress and goals.
- Trained new hires and helped with advanced and aspects of Windchill database.
- Scheduled meetings and presented data with middle/high level management.

PDM Analyst
Ryzen Solutions for Velodyne Lidar, Morgan Hill, CA **Jul 2014 - Jan 2015**

- Used Arena PLM database to import parts, create ECOs and BOMs, submit changes, edit vendor/manufacturer sourcing, create items and add files.
- Identified components and values to update in engineering schematics.
- Produced drill drawings for PCBs showing drill patterns and tolerances.
- Synchronized engineers' component libraries with Velodyne part numbers/properties/attributes by combining schematics and Arena data.

Production Manager
Diablo Valley Signs, Concord, CA **Jun 2010 - May 2013**

- Established relationships, communicated and negotiated with people of various level of education and work experience (engineers, general contractors, architects, city inspectors, new hires, trainees, external and internal office staff).
- Performed office duties including handling multi-phone lines, process forms, permit requests, estimates and schedule phases of installation.
- Responsible for installation/troubleshooting of LED and Neon Channel Letter signs.
- Utilized knowledge of Neon Transformers at 12, 000V-15, 000V, Photocells, Timers(digital/analog), raceways, conduit pipe (installation and wire pulling), MC Cable, .
- Experienced with operating computer interfaced CNC router software and hardware.